

IMPACT ENERGY

Energy cannot be created or destroyed, but it can be converted into different forms. When two objects slam together at high speed, the energy of their speed is turned into heat. This is why big meteorites create impact craters surrounded by pieces of rock that have been melted. The Barringer Crater in Arizona was created by a meteorite 165 ft (50 m) wide—the impact energy was converted into enough heat to melt the rock.



MELTDOWN

As space rocks crashed into the early Earth, the impacts generated so much heat that the whole planet was kept hot and molten. Gravity dragged most of the heaviest ingredients of the molten rocks towards the center, creating a metallic core surrounded by layers of lighter, less metallic rock.



PLANET EARTH



NUCLEAR FURNACE

As the rate of accretion slowed, Earth started to cool and solidify again, this time in layers surrounded by the cool, brittle crust. However, radioactivity within the interior of the planet keeps it hot, and this drives the geological processes that generate earthquakes and volcanoes.



◀ **RED-HOT LAVA**
Radioactive rocks deep inside Earth produce nuclear energy that is transformed into heat. This can melt solid rock and turn it into volcanic lava, like this on Hawaii.

LETTING OFF STEAM

As the crust was forming, gas erupting from huge volcanoes formed Earth's early atmosphere. The volcanic gas included vast amounts of water vapor that created enormous clouds. Over millions of years, torrential rain poured down to fill the first ocean, which probably covered the entire planet.

